

Homework 1: Fast-Food Nutrition

Descriptive statistics and data manipulation

Purpose: Demonstrate that you can inspect, filter, sort, summarize, and communicate results from a real dataset.

Data and submission requirements

- Use the supplied Excel file: Fast foods - Nutritional information.xlsx.
- Try to generate results programmatically; do not type calculated values into tables by hand.
- Round displayed means and standard deviations to two decimal places unless a question says otherwise.

Question 1 — Understand the dataset

Describe the dataset in a compact summary. Include the number of observations and variables, the number of unique restaurants and food types, the data type of each variable, missing-value counts, and appropriate descriptive statistics for serving size, calories, sodium, and sugars. Present your results clearly, then write 2–3 sentences describing what you learned.

Question 2 — Find extreme observations

Identify the five menu items with the highest sodium content. Report restaurant, item, food type, sodium, and calories, sorted from highest to lowest sodium. Then identify the minimum- and maximum-calorie menu items and report all eight variables for each. In 2–3 sentences, explain why examining complete records is more informative than reporting only the extreme values.

Question 3 — Filter with multiple conditions

Create a subset containing menu items with fewer than 500 calories AND less than 1,000 mg of sodium. Report the number of qualifying items. Then count qualifying items by restaurant and sort the counts from highest to lowest. If restaurants tie, report all tied restaurants. Briefly explain exactly what each row in your restaurant summary represents.

Question 4 — Summarize by food type

For each food type, calculate the number of menu items and the mean, standard deviation, minimum, and maximum of calories. Sort the table from highest to lowest mean calories. Write two observations supported by values in your table. Use the sample standard deviation used by your software's standard summary function.

Question 5 — Summarize by restaurant

For each restaurant, calculate the number of menu items, mean calories, mean sodium, and mean sugars. Sort first by mean calories from highest to lowest. Then identify which restaurant is highest and lowest on each mean nutritional measure. In 3–4 sentences, explain why these restaurant summaries should not automatically be interpreted as comparisons of otherwise equivalent menus.

Question 6 — Use two grouping variables

Calculate mean calories for every restaurant × food-type combination that appears in the dataset. For each food type, identify the restaurant with the lowest mean calories. Return a six-row table with food type, restaurant, mean calories, and the number of items used to calculate that mean. State how your procedure would handle ties, even if no tie occurs.

Question 7 — Debug an analysis

The following code is intended to identify the restaurant with the highest mean calories, but it returns the wrong restaurant. Explain the error, correct the code, and verify the answer with the corrected result.

Python:

```
df.groupby('restaurant')['calories'].mean().sort_values().head(1)
```

R:

```
df |> group_by(restaurant) |> summarise(mean_calories = mean(calories)) |> arrange(mean_calories) |> slice_head(n = 1)
```